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Similarity network approach on transcriptomic data of patients with systemic infectious diseases.

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Introduction

The high complexity in patients with severe infectious diseases, caused by bacterial, fungal and also viral pathogens (COVID-19), have imposed a challenge to design new strategies of classified patients, through identification of specific molecular variation on different "omics" levels (DEG in single cells, SNPs, ect.). In fact, complex and partially unknown infectious diseases needs new approach, combining data from patient samples, especially in pandemic context. Similarity Network Function (SNF) is part of explorative methods that look at for precision medicine (Pai and Barden, 2018). This new point of view allows to highlight new aspects of complex disease with potentially fatal outcome, so much that it has given an important support to diagnosis in aggressive types of bone marrow cancers (Bhalla et al., 2022). SNF workflow is based on designing of a similarity network among patients for each data type and integration of these networks into one single similarity network using a nonlinear combination method (Wang et al. 2014). In this experience SNF approach was applied to create PBMCs samples network differentially stimulated in order to mirror the early phases of systemic infection with potentially fatal outcome. Here, we propose the application of SNF based of PBMCs of healthy subjects, exposed for 24 hours to three important pathogens, involved also in systemic infectious diseases, *Pseudomonas aeruginosa*, *Candida albicans* and *Mycobacterium tuberculosis.*
The analysis was fosused on single cell RNAseq in specific population of PBMCs (in particular, CD4 and CD8 T cells, Monocytes and NK) and microbial data, reported in 1M-scBloodNL project (Oelen et al. 2022).

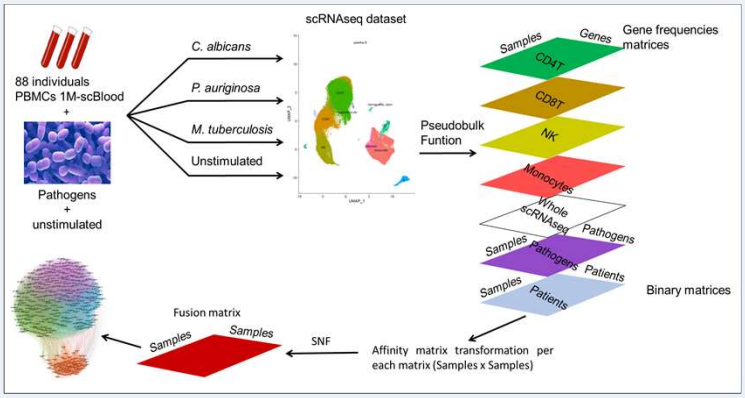

Figure 1: Similarity Network workflow on scRNAseq and microbial data for identification of PBMCs exposed to pathogens and unstimulated.

	P. auriginosa	M. tuberculosis	C. albicans	Unstimulated
Cluster_1	8	45	48	0
Cluster_2	0	0	0	70
Cluster_3	37	18	8	1
Cluster_4	26	8	15	0

Table 1: PBMCs samples exposed to three pathogens and unstimolated assigned to four clusters.

Materials and Methods

The data for this analysis is composed by gene expression data of PBMCs, collected from 88 healthy subject and sequenced with Chromium single-cell RNA-seq chemistry V2 with 625677 cells with 23269 genes, downloaded from 1M-scBloodNL (https://www.eqtlgen.org/sc/datasets/1m-scbloodnl.html). For each subjects, the PBMCs were exposed for 24 hours to three pathogens, which could be involved in systemic infectious diseases, *Pseudomonas aeruginosa, Candida albicans* and *Mycobacterium tuberculosis*. To test this approach, the 1M-scBloodNL scRNAseq database was chosen to reduce the number of variables uncosiderable, such as ICU condition, treatments and time from onset. The whole scRNAseq dataset was handled used R packages Seurat V3. From original database, PBMCs exposed to three patogens for 24 hours, along with one sample unstimulated, of 71 subjects were filtered and chosen for this analysis. We obtained four *.rds* files, belonged to following categories: *C. albicans* (CA24h), *M. tubericulosis* (MTB24h), *P. aeruginosa* (PA24h) and unstimulated (UT). Each scRNAseq dataset were scaled and analyzed for identify cells types. The first 30 PCs for principal-component analysis (PCA) were used, while UMAP was performed on the top 30 PCs for visualizing the cells, wich 0.5 resolution. CD4 and CD8 T cells, Monocytes and NK was filtered from other celltypes, and gene-barcode matrices were converted in gene expression data of bulk RNAseq (pseudobulk) by *aggregatedata* function of SNFtools R package (Wang et al., 2014). The whole scRNAseq data was aggregate to report all information. Two binary matrices with pathogen and patient information were added to the clustering. The matrices were transformed in similarity matrices (Samples x Samples) and finally every matrices were fused by *SNF* function, obtaining a fused affinity matrix that allow to group PBMCs samples. The *calNMI* function allowed to calculate the mutual information between matrices, quantifing the overlapping degree of each single matrix to the fused matrix, estimating also the contribution to the clustering.

Results

Firstly, the SNF has allowed to find groups of PBMC samples, basing on information within seven matrices. The SNF allowed to identify four sample groups, in which the unstimulated PBMC samples showed a high distance to samples stimulated. Among exposed samples, three clusters were also identified, even if a clear separation each others were not shown. Such information falls into the gene expression results in original analysis, where few number of unique genes were as overexpressed for specific pathogen exposition (10.3% of all DEGs) (Oelen et al. 2022). However, the significant correlation between clusters and pathogens was observed (Pearson's Chi-squared test X-squared = 358.63, df = 9, p-value < 2.2e-16), while it was not highlighted for patients. Finally, network model based on fused matrix showed clearly the classification of PBMCs samples.
The main contribution to the fused matrix (NMI) highlighted the high overlapping of pathogens and whole bulk scRNAseq matrix (NMI >0.45). As regarding celltypes, monocytes showed a moderate contribution to the final representation (NMI >0.2). Such data was also interpretable with early activation of innate immunity in monocytes, after the invasion of microbes in the blood stream. The patient variable seems to be few overlapping with diversity described in fused matrix, suggesting this cell response could be not linked to the specific response of patients.

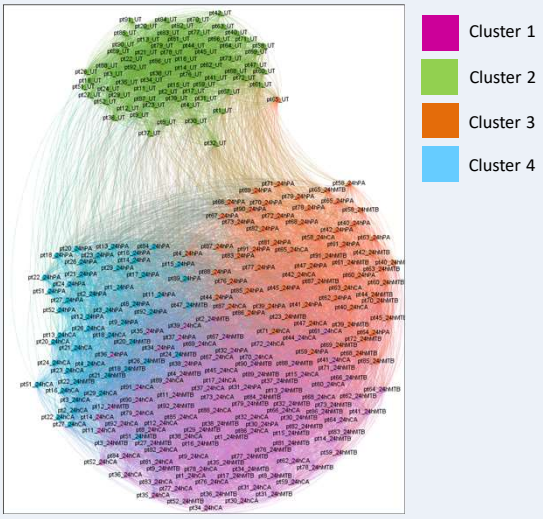

Figure 2. Network model based on fused matrix that showed the cluster of PBMCs samples unstimulated and stimulated with pathogens.

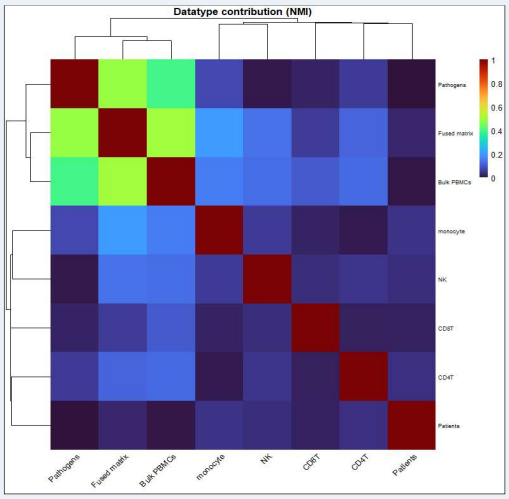

Figure 3. Heatmap of NMI values, which quantifies the contribution of each data matrix to fused matrix. The matrices tested were CD4T, CD8T, NK, Monocyte, whole Bulk scRNAseq, pathogens and patients. Values close to 1 (color red) means high similarity and high contribution to fused matrix.

Conclusions

This experience enabled to determine distinct subgroups of PBMCs samples, caused by molecular variation in transcriptomic data per single cells, through the application of SNF on PBMCs exposed to pathogens well-known in controlled conditions. This preliminary analysis will allow to design Patient Similarity Network of systemic infectious diseases, in which, integrating large database of clinically informative, patient clusters with different disease severity degree might be defined. In conclusion, this approach could provide a diagnostic approach for precision medicine in infectious diseases and then response quickly to future pandemic caused by unknown pathogens. However, for this to be applied, it will need to improve molecular resolution power in systemic infectious diseases, adding other "omics" data in both human host (eQTL, whole exome, blood proteomics, etc.) and pathogen (WGS, AMR and virulence genomic profiling).

References: Pai S, Bader GD. Patient Similarity Networks for Precision Medicine. J Mol Biol. 2018;430; Bhalla et al. Patient similarity network of newly diagnosed multiple myeloma identifies patient subgroups with distinct genetic features and clinical implications. Sci Adv. 2021;7(47):eabg9551.; B. Wang,et al., Goldenberg, Similarity network fusion for aggregating data types on a genomic scale. Nat. Methods 11, 333–337 (2014); Oelen R, et al. Single-cell RNA-sequencing of peripheral blood mononuclear cells reveals widespread, context-specific gene expression regulation upon pathogenic exposure. Nat Commun. 2022;13(1):3267. As regarding 1M-scBloodNL project, we thanks the participants and the staff of Lifelines-DEEP DAG2+ for their collaboration. Funding for the project was provided by the ERC Starting Grant #637640awarded to prof. Dr. L. Franke.